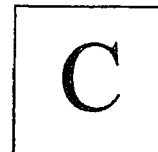


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B.Tech. Degree VII Semester Supplementary Examination May 2024

ME 19-205-703 MACHINE DESIGN II (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Understand the customers need, formulate the problem and draw the design specifications.
 CO2: Understand component behaviour subjected to loads and identify the failure criteria.
 CO3: Apply the concepts of stress analysis, theories of failure and material science to analyze, design and/or select commonly used machine components.
 CO4: Analyze the stresses and strains induced in a machine element.
 CO5: Design machine components using theories of failure.
 CO6: Get basic knowledge of the preparation of working drawings by including the manufacturing details like tolerance, surface finish etc.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

(Use of Design Data Handbook is permitted)

(Data not given may be suitably assumed. All such assumptions must be clearly stated)

PART A

(Answer **ALL** questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	With a simple diagram explain the working of centrifugal clutch.	3	L2	1	1
(b)	What are the various factors that decide the selection of belt drives? Explain.	3	L2	1	1
(c)	Write down the assumptions considered for deciding the type of gear for a particular application.	3	L2	2	1
(d)	Explain the phenomenon of interference in involute gears and the conditions to be satisfied in order to avoid interference.	3	L2	2	1
(e)	Why is hydrodynamic journal bearing called self acting bearing?	3	L2	2	1
(f)	Differentiate between median life and rating life of a bearing.	3	L2	2	1
(g)	Explain the design recommendations for rolled steel sections.	3	L2	1	1
(h)	Discuss the economic and product design considerations in machining.	3	L2	2	1

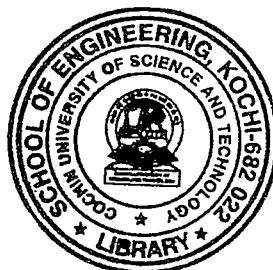
PART B

(4 × 12 = 48)

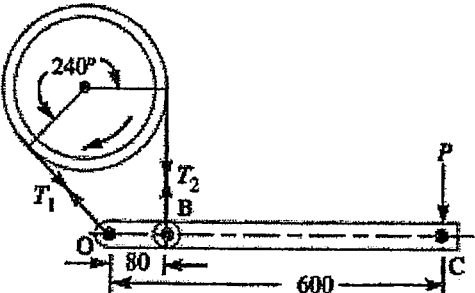
II.	A leather faced cone clutch transmits power at 50 rpm. The semi cone angle is 12.5°. The mean diameter of the clutch is 300 mm, while the face width of the contacting surface of the friction lining is 100 mm. The coefficient of friction is 0.2 and the maximum intensity of pressure is limited to 0.07 N/mm ² . Calculate the force to engage the clutch and the power transmitting capacity.	12	L3	5	2
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OR

(P.T.O.)



BTS-VII(S)-05-24-2374

		Marks	BL	CO	PO
III.	The drum of a simple band brake is 450 mm. The band embraces $3/4^{\text{th}}$ of the circumference of the drum. One end of the band is attached to the fulcrum pin and the other end is attached to a pin B as shown in figure. The band is to be lined with asbestos fabric having a coefficient of friction 0.3. The allowable bearing pressure for the brake lining is 0.21 N/mm^2. Design the band, shaft, key, lever and fulcrum pin. The material of these parts is mild steel having permissible stress as follows. $\sigma_t = \sigma_c = 70 \text{ MPa}$, $\tau = 56 \text{ MPa}$  All dimensions in mm.	12	L3	3	2
IV. (a)	A gear drive is required to transmit a maximum power of 22.5 kW. The velocity ratio is 1:2 and r.p.m of the pinion is 200. The approximate centre distance between the shafts may be taken as 600 mm. The teeth have 20° stub involute profiles. The static stress for the gear material (which is cast iron) may be taken as 60 MPa and face width as 10 times the module. Find the module, face width and number of teeth on each gear.	6	L3	3	2
(b)	A pair of 20° full depth involute teeth bevel gears connect two shafts at right angles having velocity ratio 3:1. The gear is made of cast steel having allowable static stress as 70 MPa and the pinion is of steel with allowable static stress as 100 MPa. The pinion transmits 37.5 kW at 750 r.p.m. Determine module and face width.	6	L3	3	2
OR					
V. (a)	A pair of helical gears with 30° helix angle is used to transmit 15 kW at 10000 r.p.m. of the pinion. The velocity ratio is 4:1. Both the gears are to be made of hardened steel of static strength 100 N/mm^2. The gears are 20° stub and the pinion is to have 24 teeth. The face width may be taken as 14 times the module. Find the module and face width from the standpoint of strength.	6	L3	4	2
(b)	Find the dimensions of a worm gear drive which is used to transmit 5 kW from a worm running at 1120 rpm to a worm wheel rotating at 70 rpm. Select hardened steel for worm and manganese bronze for gear. The pressure angle is 20 degree Full Depth Involute.	6	L3	4	2

(Continued)

BTS-VII(S)-05-24-2374

		Marks	BL	CO	PO
VI.	(a) What are rolling contact bearings? Discuss their advantages over sliding contact bearings.	6	L2	2	1
	(b) A single-row deep groove ball bearing is subjected to a radial force of 8 kN and a thrust force of 3 kN. The shaft rotates at 1200 rpm. The expected life L_{10h} of the bearing is 20000 hours. The minimum acceptable diameter of the shaft is 75 mm. Select a suitable ball bearing for this application.	6	L3	3	2
OR					
VII.	The following data is given for a 360° hydrodynamic bearing: journal diameter = 100 mm, bearing length = 100 mm, radial load = 50 kN, journal speed = 1440 rpm, radial clearance = 0.12 mm, viscosity of lubricant = 16 cP, Calculate: (i) Minimum film thickness. (ii) Coefficient of friction. (iii) Power lost in friction.	12	L3	3	2
VIII.	Write the general design recommendations for (i) Casting process. (ii) Forging process.	12	L2	2	1
OR					
IX.	(a) What are the major guidelines for assembly improvement?	6	L2	6	1
	(b) Discuss the forging process and briefly explain their classifications.	6	L2	2	1

Bloom's Taxonomy Levels

L2 - 45%, L3 - 55%.

